Marco Adeleye

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SKILLS SUMMARY

- **Programming:** C, C++, C#, Python, Java, Swift, JavaScript, ROS2, React, Node.js, NumPy, Pandas, Pytest
- **Development Tools:** Git, GitHub, Linux, Docker, Xcode, Insomnia, Jira
- **Embedded & Systems:** NVIDIA Jetson, Raspberry Pi, CAN Bus, FPGA, VHDL, Verilog

WORK EXPERIENCE

Business Analyst *January 2026 – August 2026*

The Reynolds and Reynolds Company, Mississauga, ON

- Worked in a team of 6+ Business Analysts to perform Quality Assurance and verify Software Developer code changes.
- Maintained constant communication with 5+ Software Developers to ensure changes reflect customer needs.
- Developed and applied 15+ test cases per change via Insomnia, ensuring API functionality and preventing regressions.
- Managed and continuously updated Kanban Tickets/Documentation on GitHub with Quality Assurance progress.
- Reduced overall regression and potential errors in Software Developer changes by 83%.

Jr. Software Developer *May 2025 – December 2025*

TealBook, Toronto, ON

- Worked in a team of 4 members to improve the company's legal entity matcher XGBoost ML model.
- Created features to improve the AI model's matching capabilities for entity names, addresses, and jurisdiction.
- Created and reviewed over 11,000 mock candidates for training data via Pandas.
- Maintained constant communication and feedback loops with product and app development teams.
- Developed and evaluated XGBoost model features, contributing to F1-score improvement from 60% to 90%.

ENGINEERING PROJECTS

Wireless Network Integration on Mars Rover *September 2025 – August 2026*

Canadian International Rover Challenge, Drumheller, ON

- Implemented remote Rover control via Joystick with live motor data and simulated Rover physics in Webots.
- Developed a Base Station via React, Python, ROS2, and NiceGUI visualizing rover status, telemetry, and camera feeds.
- Configured Ubiquiti Rocket M2 wireless antenna network to establish wireless communication with a range of 30 km.
- Made use of CANBUS firmware to integrate ODrive S1 motors and Dynamixels with the ROS2 network.
- Developed several team Docker images to manage package libraries and dependencies on Jetson Orin Nano.
- Collaborated within a multidisciplinary team of 20+ members as Software Team Lead across software, firmware, electrical, and mechanical disciplines to integrate Rover systems during competition.

Large-Scale Software Architecture Team Project (NLTK Open-Source Contribution) *September 2025 – December 2025*

University of Guelph, Guelph, ON

- Worked in a 7-member team enhancing the Natural Language Toolkit (NLTK), a globally used open-source NLP library.
- Implemented a robust fix for missing tokenizer resources, ensuring backward compatibility within a large codebase.
- Developed Python-based, object-oriented solutions, updated packaging and installation workflows, and authored 100+ Pytest unit and regression tests with 100% coverage to prevent regressions.
- Collaborated via GitHub (feature branches, pull requests, merge conflict resolution) and Jira (Agile/SCRUM).
- Delivered a solution enabling users to install required tokenizers directly via pip while preserving user data integrity.
- Successfully implemented changes via a pull request contributing to NLTK.

EDUCATION

Bachelor of Engineering, Computer Engineering (Co-op) *September 2023 – May 2028*

University of Guelph, Guelph, ON

- **Concentrations:** Robotics & Software Design.
- **Related Coursework:** Autonomous Systems & Robotics, Software Systems Development and Integration, Embedded Reconfigurable Systems, Wireless Sensor Networks, Large-Scale Software Architecture, Data Structures.
- On the College of Engineering Dean's Honour list.